Drought severity in a constrained circulation ensemble of the 2020 Southwest United States drought

D. Kennedy[1,2], D. M. Lawrence[1], I. R. Simpson[1]

[1]Climate and Global Dynamics Laboratory, NSF-NCAR
[2]Earth Research Institute, University of California Santa Barbara

Key Points:

- A constrained circulation ensemble can reproduce the 2020 Southwest United States drought.
- Soil water initialization influences drought severity (soil moisture, runoff, temperature extremes), but does not influence the precipitation.
- Warming from climate change may have exacerbated the monsoon failure, but to a small extent.
- The constrained 2020 drought and an 1850 analog are comparable to extreme droughts in the free running model, but a 2090 analog would be exceptionally dry.

Corresponding author: Daniel Kennedy, `djk2120@ucar.edu`

Abstract

TKTK

Plain Language Summary

TKTK

1 Introduction

Water in the Southwest United States (SWUS) is scarce, with myriad competing uses including agriculture, public water supply, recreation, and supporting the local ecology (Dettinger et al., 2015). The abundance of water is also related to fire risk and temperature extremes (Holden et al., 2025; Hirschi et al., 2011). Climate change has added strain to the water supply via reduced snowpack, increased atmospheric demand, and increased precipitation variability (Wieder et al., 2022; Albano et al., 2022; Pendergrass et al., 2017). The summer of 2020 was the driest of the meteorological record within the SWUS (Mankin et al., 2021; Hoell et al., 2022) and arrived in the midst of a 22-year megadrought (2000-2021), which was drier than any other extended drought in a soil moisture reconstruction dating back to 800 C.E. (Williams et al., 2022). Consequences of this drought (and its continuation into 2021) included severe water restrictions, billions in economic damages, and a staggering fire season (Higuera & Abatzoglou, 2021; Jiang et al., 2025)

Drought in the SWUS, and its intersection with climate change, is challenging to understand from observations alone, and has been frequently investigated with the aid of global or regional climate models (Cook et al., 2018). Models tend to agree on trends in temperature (Kumar et al., 2013; Papalexiou et al., 2020), with an associated increase in vapor pressure deficit and reduction in surface soil moisture (Fang et al., 2022; Berg et al., 2017). The forced responses of precipitation and runoff are more heterogeneous and more uncertain (Kumar et al., 2013; Shepherd, 2014; Wu et al., 2024). Precipitation is difficult to predict across the SWUS, due to complex orography and the dynamic nature of the North American Monsoon (Cook et al., 2020; Pascale et al., 2017). There is also evidence that models may systematically underestimate the drying effects of climate change on near-surface atmospheric vapor pressure in semi-arid regions (Simpson et al., 2024). As such, it can be difficult to make strong inference regarding precipitation variations or trends in the region using large-scale coupled climate models. Despite the recent megadrought, the only forced precipitation trend identified for the SWUS at this point is a small decrease during the spring season (Seager et al., 2022). The general consensus seems to be that climate change has indeed exacerbated SWUS drought in a general sense, but that the precipitation shortfalls of the extreme 2020 drought themselves are primarily attributable to random (natural) atmospheric variations, with potential contributions from a shift in the Pacific Decadal Oscillation from the positive phase (most of the 1980s and 1990s) to its negative phase (most of the 2000s and 2010s) (Mankin et al., 2021; Seager et al., 2022; Jiang et al., 2025).

Climate model large ensembles are often used to discern forced responses from natural climate variability (Deser et al., 2020). The CESM2 large ensemble (Rodgers et al., 2021) provides 100 ensemble members with varied initial conditions, yielding 100 distinct realizations of the historical period, and a large sample size for investigating drought dynamics in the SWUS. Instead of one instance of the year 2020, there are 100, such that we can generate large samples of SWUS summers, in order to better analyze drought dynamics in the context of natural variability. However, the statistical basis can still become small when focusing explicitly on extreme quantiles, or when filtering for specific circulation criteria. Attribution of the 2020 SWUS drought is challenging, partly because of our imperfect models, but also because we are trying to diagnose changes not in the mean behavior, but in the severity of a rare event. Climate hazards are often sporadic,

with risks that are disproportionately concentrated in the extreme quantiles of the probability distributions, e.g. "the hundred-year flood" (Newman & Noy, 2023). As such, to effectively characterize risk using climate models, especially given finite computing resources, it can be helpful to design experiments that up-sample these hazard occurrences.

One such technique is to run land-atmosphere coupled simulations forced with observation-based SSTs from an extreme event to understand the association of SST patterns and antecedent conditions with hazard risk (Hoell et al., 2022; Seager et al., 2022). However, if the SSTs are not directly implicated in drought risk, such ensembles will not meaningfully upsample drought events, as seems to be the case with the 2020 summer drought in the SWUS (Jiang et al., 2025). Another technique, reinitialized ensemble boosting, involves branching a fully coupled simulation ahead of a key model-based event, and introducing roundoff level atmospheric perturbations to generate an ensemble of short-duration, fully-coupled simulations. This has been used to better understand the proximate drivers of heat extremes, and to demonstrate the potential of even more extreme heatwaves than seen in the observational record (Gessner et al., 2021; Fischer et al., 2023). This technique is very effective at simulating extreme conditions, but will generally span only a small range of initial conditions, due to the shared initialization among ensemble members.

In this study we impose a constrained circulation ensemble approach, where we not only force with observation-based SSTs, but also nudge winds to reanalysis to capture the specific circulation patterns of the 2020 SWUS drought. This allows us to generate an ensemble with drought in every simulation, but that also spans a wide range of land initializations. This approach was effectively used to investigate the influence of soil moisture preconditioning on the 2021 Pacific Northwest heatwave (Duan et al., 2025). Using this framework, along with an existing fully coupled large ensemble, we are able to ask:

1. Can a climate model reproduce the 2020 SWUS drought when forced with the observed large-scale circulation?
2. How do antecedent conditions, specifically soil moisture, influence drought severity?
3. To what extent is the 2020 SWUS drought attributable to unforced climate variability, and should we expect future climate change to exacerbate the impacts of a drought-inducing circulation pattern like that which occurred in 2020?

2 Methods

2.1 Model description

For this study we utilized the Community Earth System Model, version 2 (CESM2) (Danabasoglu et al., 2020). We ran a suite of simulations in land-atmosphere coupled mode (namely the FHIST_BGC component set) using a data ocean (simulations described in Section 2.3). We compared our simulations to the CESM2 Large Ensemble (CESM2-LE) (Rodgers et al., 2021), which consists of 100 transient simulations (1850-2100) of the fully coupled configuration of CESM2. The simulations follow historical forcing through 2014 and then switch to the SSP3-7.0 emissions scenario. The large ensemble is useful, because it provides a large number of model realizations, sampling across a range of internal climate variability states. For the sake of comparison, we utilized the CESM2-LE codebase for our simulations and the same forcing where applicable.

2.2 Data

We opted to force our simulations with the NOAA Extended Reconstructed Sea Surface Temperature V5 (ERSSTv5) dataset (Huang et al., 2017). These are available

at 2° resolution, which we regridded to the CESM2 nominal 1° resolution. We utilized the wind fields from ERA5, the fifth generation European Center for Medium Range Weather Forecasts atmospheric reanalysis product (Hersbach et al., 2020), in order to nudge our simulations to the 2020 circulation (described further in Section 2.3). All other forcings (e.g. aerosols, CO_2, land use) follow the protocol of the CESM2-LE (Rodgers et al., 2021), including emissions from CMIP6, that follow the historical and then the SSP3-7.0 emissions pathway (Eyring et al., 2016). We compared our simulations to the Global Precipitation Climatology Project (GPCP) Monthly Analysis Product (Adler et al., 2003). These data are available at 2.5° resolution, which we likewise regridded to the CESM2 nominal 1° resolution.

2.3 Experimental design

Our study is focused on the 'four corners' regions of the Southwest United States, which is defined as the region encompassing four adjoining states: Colorado, Utah, New Mexico, and Arizona. This region was chosen based on the spatial footprint of the 2020 drought (Supp Figure 1), as well as its significance for water management (Mankin et al., 2021). Our goal was to create a set of simulations with extreme drought conditions, while still allowing for local land-atmosphere coupling, in order to investigate the influences of antecedent conditions and climate change.

We ran multiple ensembles of simulations using CESM2, each consisting of 30 members, with varied initial conditions. For each simulation, winds were nudged to the ERA5 reanalysis for the year 2020. Nudging occurred four times per day in each grid cell outside the box bounded by 28°N, 47°N, 130°W, 96°W (red in Figure 1a). This serves to impose the large-scale circulation associated with the 2020 SWUS drought for every ensemble member. Inside the bounding box, the atmosphere evolves freely (aside from the constraints at the box edges), allowing for local land-atmosphere coupling, which is necessary to study the effects of initial conditions. Sea surface temperatures (SSTs) were imposed from ERSSTv5, and all other forcing (e.g. CO_2, aerosols) followed the standard conventions for transient simulations as in the CESM2-LE. The simulations were initialized in April, using initial conditions from the CESM2-LE for all the requisite land and atmosphere states (e.g. temperature, soil moisture, humidity). Each simulation ran for nine months.

Our primary ensemble was based in 2020, mirroring the extreme 2020 Southwest U.S. drought. We also simulated this drought during pre-industrial and future conditions to investigate the influence of climate change. All of the ensembles were nudged to the 2020 winds, but initial conditions and other climate forcings were drawn from the appropriate time period of the CESM2-LE, which is designed to port the 2020 circulation to these two alternative climates. Initial conditions were drawn at random from the CESM2-LE from the ten year period centered on the given ensemble year (i.e. 1851-1860, 2015-2024, or 2090-2099 TK check years). SST forcing for the pre-industrial and future ensembles was anomaly-based, wherein we computed average SST anomalies for the reference period relative to 2015-2024 in the CESM2-LE and applied them to the ERSSTv5 data used in the 2020 ensemble. All other forcing (e.g. CO_2, aerosols) followed the standard conventions for transient simulations as in the CESM2-LE, which follows the SSP3-7.0 emissions scenario. The two key differences between our ensembles and the CESM2-LE are that 1) we imposed the large-scale circulation associated with the 2020 via nudging and 2) we forced the simulations with SSTs (in lieu of dynamically coupled SSTs).

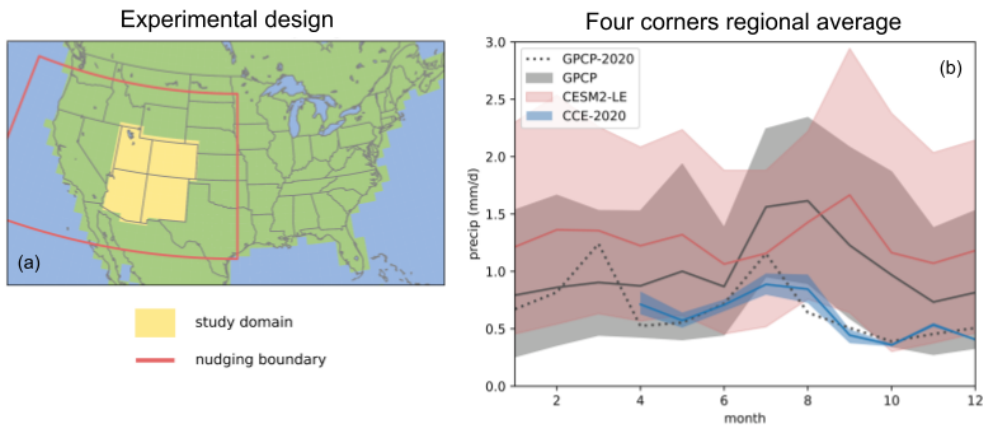

Figure 1. Map of the study domain and nudging boundary (a). All analyses were based in the 'four corners' region of the western United States, yellow area. Model simulations were nudged to reanalysis winds outside of the red box, in order to induce the observed large-scale circulation, while allowing for local land-atmosphere coupling. Precipitation climatologies (b) for the four corners region from an observations-based product (GPCP, 1981-2020) and a model large ensemble (CESM2-LE, 1981-2020), alongside our constrained circulation ensemble (CCE-2020). For each of the datasets, shading spans the 5th to 95th percentiles (across years and/or ensemble members), and the solid line tracks the mean. The dotted black line shows the reanalysis precipitation for 2020. The CESM2-LE shows a small high bias in mean and variance relative to reanalysis precipitation, and it does not accurately capture the timing or magnitude of the summer monsoon. When nudged to 2020 winds (i.e. CCE-2020), CESM2 can reproduce the 2020 drought, even though it falls outside the large ensemble envelope.

Table 1. Ensemble descriptions

Name	Winds	Climate forcing	SSTs	Initial Conditions
Control	ERA5-2020	2020 (SSP3-7.0)	ERA5-2020	CESM2-LE (2020)
Pre-industrial	ERA5-2020	1850	ERA5-2020 + CESM2-LE anomalies (1850)	CESM2-LE (1850)
Future	ERA5-2020	2090 (SSP3-7.0)	ERA5-2020 + CESM2-LE anomalies (2090)	CESM2-LE (2090)

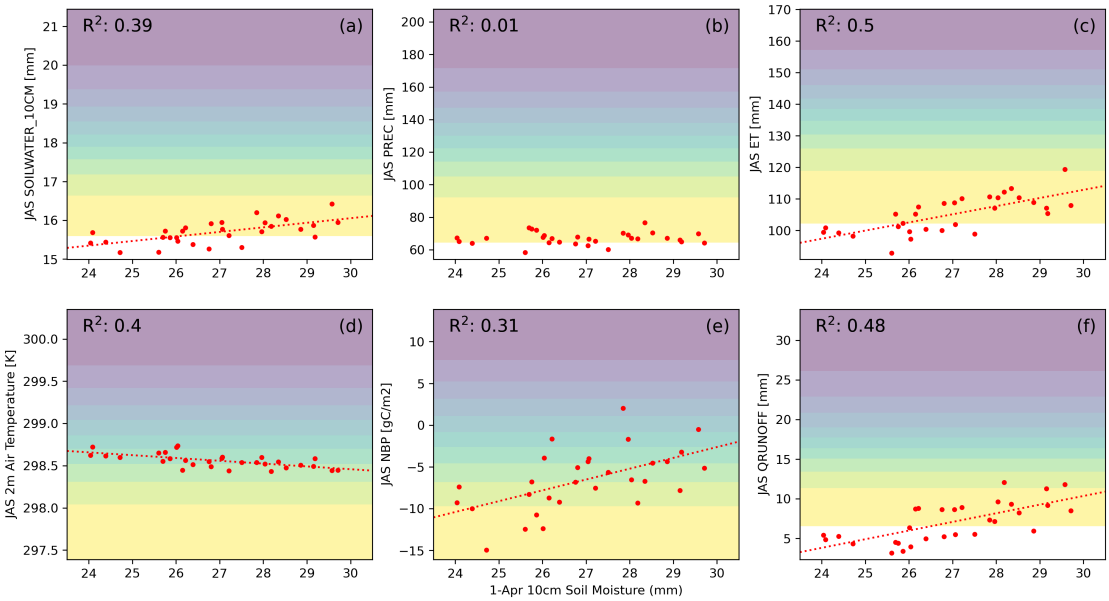

Figure 2. The influence of soil water initialization on various summer (JAS) drought indicators in the 2020 CCE, shown in the context of the variance across the CESM2-LE. In each case, the 10cm soil water content at initialization is plotted (red dots) against the summer drought indicator, with linear regression where appropriate (red lines). The colored stripes span the CESM2-LE deciles for the given drought indicator, truncated at the 1st and 99th percentiles. Dots in the white background area, exist below the 1st percentile of the CESM2-LE. Initial soil water content influences many drought indicators, but not the severity of the meteorological drought itself, i.e. summer precipitation.

3 Results and Discussion

3.1 2020 SWUS Drought

Our experiment demonstrates that CESM2 can reproduce the 2020 SWUS drought if the large-scale circulation is nudged to the observed winds (Figure 1b), as the constrained circulation ensemble (CCE) mean closely matches observed precipitation in the four corners region during 2020. This is despite an imperfect precipitation climatology in the CESM2-LE as compared to the GPCP observations across 1980-2020. The CESM2-LE slightly over-predicts precipitation mean and variance (TK, precise this) in the SWUS, compared to the GPCP reanalysis product across 1981-2020 (Figure 1b). Furthermore, CESM2 underestimates the relative contribution of summer rain to annual precipitation and does not accurately match the observed monsoon timing. When constrained to the monsoon period (JAS), precipitation statistics match more closely between the CESM2-LE and the GPCP reanalysis product (Supp Figure 3). The 2020 SWUS drought was very dry (JAS precip = 71mm, GPCP), which sits at the 1.8th percentile of the CESM2 large ensemble, primarily driven by extremely dry August and September months. Despite the extreme nature of the 2020 monsoon failure, the CCE is able to reproduce the observed meteorological drought, demonstrating an important role for the circulation pattern that occurred in producing it.

The CCE provides 30 replicates of the drought-inducing circulation, initialized across a range of soil water conditions (see Section 2.3 for details). Dry initialization exacerbates drought severity, with significant reductions in summer soil moisture, evapotran-

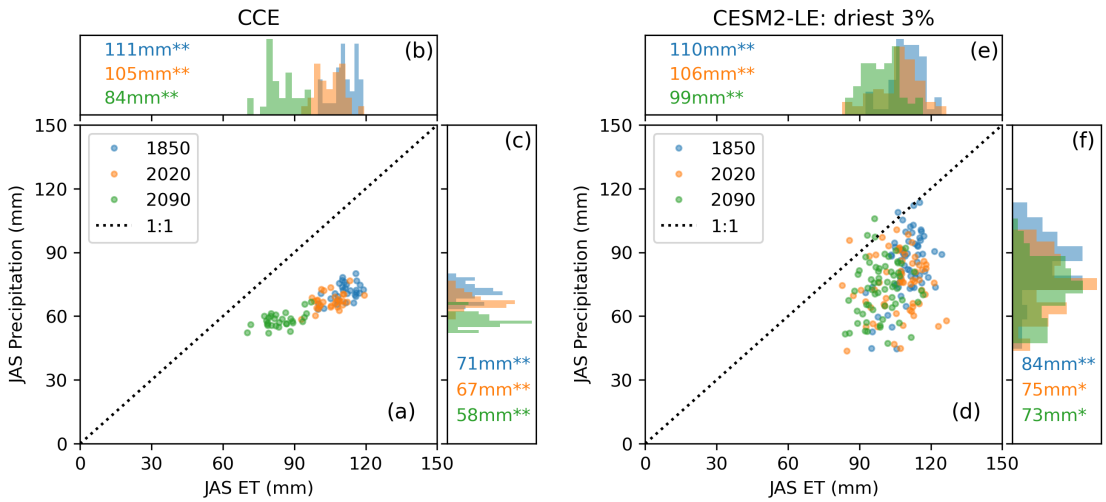

Figure 3. Summer evapotranspiration vs. precipitation in the CCEs (a) and a dry subset of the CESM2-LE (d), alongside histograms of ET (b,e) and precipitation (c,f) across three time periods. The CESM2-LE is subset to the 30 driest ensemble-years for each period from 1000 total ensemble-years (100 members, 10 years around the given year) based on JAS 10cm soil water content. Stars indicate significant differences in the mean relative to the two other time periods, via t-test at the $p<0.05$ level. Forced responses tend to reduce ET and precipitation in both the CCE and the dry subset of the CESM2-LE.

spiration (ET), runoff as well as increased temperature (Figure 2). Of the various drought indicators, ET was most closely correlated with soil water initialization (Figure 2c), but net biome production experienced the largest initialization effect (based on the number of contours crossed by the regression line, Figure 2e). Drier initial soils strongly degraded the carbon sink. Monsoon period precipitation itself was not correlated with soil water initialization, indicating that the meteorological drought severity is not sensitive to initial conditions in CESM, consistent with previous findings (Jiang et al., 2025). In line with a previous heatwave study (Duan et al., 2025), we found the CCE approach to be useful for investigating the influence of antecedent conditions on drought severity. This improves over alternative approaches that either fail to consistently impose drought (Jiang et al., 2025) or sample a smaller range of initial conditions (Gessner et al., 2021; Fischer et al., 2023).

3.2 Imposing the 2020 Circulation in alternative climates

We ran CCEs across three climate scenarios: pre-industrial, present-day, and late century in an SSP3-7.0 warming scenario. The CCEs show a climate change influence on the regional water cycle, where both ET and precipitation are reduced moving from the pre-industrial period to present-day, and again moving from present-day to the end of this century (Figure 3a-c). CCE monsoon precipitation was reduced by 4.2mm on average between the pre-industrial and present-day simulations, with another 9.0mm reduction simulated with SSP3-7.0 by the end of the century. Given that the 2020 monsoon failure represented a 64mm shortfall relative to climatological precipitation (Supp Figure 3), the CCE historical climate change effect of -4.2mm is small relative to the overall drought signal. We infer, therefore, that the meteorological drought is primarily attributable to unforced climate variability, which is consistent with prior assessments (Seager et al., 2022). As such, climate change may have exacerbated the monsoon failure, but

only slightly, and could potentially increase the severity of future precipitation shortfalls by another 9mm or roughly 10 to 15%. However, we note that the CCE experiment cannot capture any coupled circulation feedbacks that could exacerbate or mitigate the radiative response. Significant circulation changes in response to climate change forcing have been documented, but are generally less robust than processes that are primarily under thermodynamic control, such as surface temperature (Shepherd, 2014).

The reduction in monsoon precipitation observed in the CCE has not been previously observed within CESM2 (Seager et al., 2022). As such, we analyzed the CESM2-LE to see if this effect is present within the dynamically coupled version of the model. Specifically we analyzed extreme drought years; the thirty driest monsoons across 1000 simulation years, sorting by 10cm soil moisture among the 100 CESM2-LE members across the 10 years nearest our CCE. The CESM2-LE, which does include circulation feedbacks, likewise shows a drying response across the driest years, but with significantly more noise (Figure 3d-f). Because the CCE constrains the large-scale circulation, there is much less variance within the CCE than in CESM2-LE.

The response of ET to forcing in the CESM2-LE is consistent between PI to present and present to future, both showing significant reductions (Figure 3e). Precipitation declines from pre-industrial to present-day, but minimal decline occurs between present-day and the future (Figure 3f). Importantly, the forced responses at the extreme quantiles are distinct from those at moderate or wet quantiles (Supp Figure 4), highlighting the importance of investigating extremes separately from mean changes. For example, the overall CESM2-LE response to climate change in the SWUS is higher ET, but it occurs with a prominent dipole of dry years trending towards less ET, and moderate and wet years trending towards more ET.

ET behaves similarly in the CCE and the CESM2-LE, with a strong relationship between summer soil water and ET in both the CCE and the CESM2-LE (Figure 4a-c). The CCE simulations vary only by the initial conditions and are not associated with the large-scale circulation, because they were selected at random. The CESM2-LE data samples initial conditions and circulation patterns coherently, with expected covariance. Despite this, the two ensembles show comparable ET$\sim$SW relationships, indicating that the ET flux is largely determined by the soil moisture state, and is relatively agnostic to the specific circulation pattern. This is in contrast to the relationship between precipitation and ET, where the slopes of the two relationships are different (Figure 4). The CCE shows a much weaker relationship between P and ET than seen in the CESM2-LE, indicating that, constrained to the 2020 circulation pattern, adding more ET does not have a large effect on P in the model. In the CESM2-LE shifting from left to right in these plots is not only changing the soil water (as in the CCE), but is likewise traversing a range of circulation patterns from diverging to converging, with associated effects on both ET and P. The large discrepancy in the P$\sim$ET relationship between the CCE and the CESM2-LE seems to indicate that the circulation pattern has a stronger influence on precipitation than does the local water availability.

In our experiment, we ported the 2020 drought circulation to PI and future climates, which imposes the climate change radiative forcings, without any coupled changes to the circulation pattern. In many ways the CCE behaves akin to the CESM2-LE, however the CCE is much drier relative to the CESM2-LE in 2090 than it is in 1850 or 2020 (e.g., the CCE centroid is well outside the CESM2-LE contours in Figure 4). The consequences of the 2020 drought circulation are within the envelope of the CESM2-LE in the PI and 2020 experiments. However when we port this circulation to 2090, the results indicate very high values of moisture flux divergence and extremely low surface humidity, humidity levels that are significantly lower than that occur during the most extreme droughts of the CESM2-LE (Supp Figure 5). This result suggests that under late-century SSP3-7.0 conditions either the probability of this extreme circulation pattern is reduced or that coupled circulation feedbacks may partially mitigate extreme divergence events in CESM2.

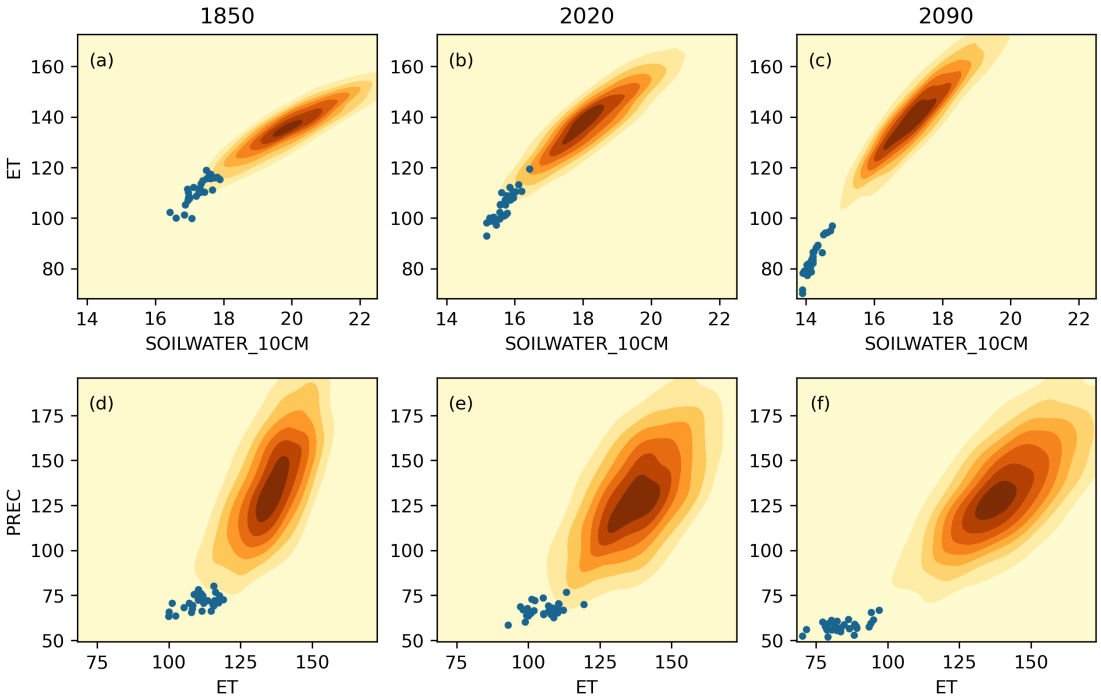

Figure 4. Summer evapotranspiration vs. 10cm soil water content from the CCEs (dots) and the CESM2-LE (scatter density contours, deeper reds indicate more density) across the three time periods. The relationship in the CCEs closely follows the CESM2-LE pattern. The CCEs exists at the dry extreme of the CESM2-LE, or even beyond, as in the 2090 scenario.

4 Conclusions

This study highlights the utility of constrained circulation ensemble experiments to investigate climate extremes. Our 2020 CCE is able to reproduce the 2020 SWUS drought, despite known biases in the fully coupled CESM2. This experiment demonstrates that antecedent soil moisture exacerbates drought severity, but not the precipitation deficits. We likewise found a small, but robust, influence of the thermodynamic impacts on climate change on precipitation amount during the 2020 drought in the CCE. That is, despite constraining circulation to be the same in each case, we found that the impacts of that circulation pattern on precipitation was to enhance the associated precipitation deficits. An even larger negative effect on precipitation is found by the end of the 21st century. Whether such an impact would actually manifest is unclear given the distinction between the behavior of the CCE and the free running CESM2-LE by the end of the 21st century which raises the possibility that forced circulation change and/or circulation feedbacks may limit the occurrence of such extreme drought inducing circulation patterns .

The 2020 SWUS drought analogues herein impose several climate change effects (e.g. SST warming, radiative forcing, aerosol changes). However we are repeating the same circulation across all three time periods, such that if the 2020 drought circulation is rendered unattainable by some forced response, inference in the constrained system would not apply to the fully coupled system. We found some evidence of this in our experiments, where, for certain effects, the CCE showed stronger responses to climate change than the fully coupled CESM2-LE. As such, whereas there may have been a small contribution of climate change to the historic precipitation shortfall of 2020 ($\sim$6% in the CCE), it is not clear that future climate change would further exacerbate monsoon failure in the SWUS. While the 1850 and 2020 ensembles looked comparable, in 2090 the CCE is outside of the range found in the CESM2-LE simulation with regard to 2m vapor pressure and 10cm soil moisture. This may indicate a propensity for CESM2 to mitigate extreme moisture flux divergence via circulation feedbacks, but would need to be investigated more specifically. This could be of particular interest, given that fully coupled Earth system models show erroneous trends in near surface atmospheric vapor pressure (Simpson et al., 2024).

We have higher confidence in our inference with regard to investigating preconditioning effects using the CCE approach. Drier initializations were associated with lower ET, lower runoff, higher temperatures, and a weaker carbon sink, serving to significantly exacerbate the drought. However, initialization did not have a strong influence on precipitation, itself. Precipitation in our analyses was generally controlled by the circulation pattern, more than by local water availability. The CCE approach is particularly useful for investigating the influence of antecedent conditions, because we are able to force the meteorological drought in all simulations, but also sample a wide range of initial conditions.

Because climate risk tends to be concentrated in extreme events, it is prudent to design experiments to study such events beyond their natural occurrence rate in free-running simulations. Likewise forced changes to the mean in any given variable may not transpose directly to the extreme quantiles. For example, we found that for CESM2, climate change tends to increase ET in our study domain on average, but actually results in a decrease in ET during drought years. Frameworks to better resolve extreme events include initial condition large ensembles, AMIP ensembles, branched reinitialization ensemble boosting, and the constrained circulation ensemble. We find the CCE framework to be a useful tool for investigating extreme events, particularly when the simulations can be analyzed relative to an unconstrained initial condition large ensemble.

Open Research Section

We will upload the CCE data to a DOI-minted repository, after the first manuscript review. For the time being, we have uploaded our post-processed data to a temporary location online (`https://webext.cgd.ucar.edu/FHIST_BGC/swus.cce.2020/`). Data from the GPCP and CESM2-LE are openly available online, but we have likewise uploaded our post-processed data for the four corners region to that website, for ease of review. Scripts to set up our simulations, post-process the data, and execute our analyses are online at `https://github.com/djk2120/mapp_droughts`.

Acknowledgments

DK, IRS, and DML acknowledge funding from the National Oceanic and Atmospheric Administration Modeling, Analysis, Predictions and Projections program (award number NA20OAR4310413) and funding from the NSF National Center for Atmospheric Research, which is a major facility sponsored by the NSF under Cooperative Agreement No. 1852977 Global Precipitation Climatology Project (GPCP) Monthly Analysis Product data provided by the NOAA PSL, Boulder, Colorado, USA, from their website at https://psl.noaa.gov

References

Adler, R. F., Huffman, G. J., Chang, A., Ferraro, R., Xie, P.-P., Janowiak, J., ... Nelkin, E. (2003). The Version-2 Global Precipitation Climatology Project (GPCP) monthly precipitation analysis (1979–present). *Journal of Hydrometeorology*, *4*(6), 1147 - 1167. Retrieved from `https://journals.ametsoc.org/view/journals/hydr/4/6/1525-7541_2003_004_1147_tvgpcp_2_0_co_2.xml` doi: 10.1175/1525-7541(2003)004⟨1147:TVGPCP⟩2.0.CO;2

Albano, C. M., Abatzoglou, J. T., McEvoy, D. J., Huntington, J. L., Morton, C. G., Dettinger, M. D., & Ott, T. J. (2022). A multidataset assessment of climatic drivers and uncertainties of recent trends in evaporative demand across the continental united states. *Journal of Hydrometeorology*, *23*(4), 505 - 519. Retrieved from `https://journals.ametsoc.org/view/journals/hydr/23/4/JHM-D-21-0163.1.xml` doi: 10.1175/JHM-D-21-0163.1

Berg, A., Sheffield, J., & Milly, P. C. D. (2017). Divergent surface and total soil moisture projections under global warming. *Geophysical Research Letters*, *44*(1), 236-244. doi: 10.1002/2016GL071921

Cook, B. I., Mankin, J. S., & Anchukaitis, K. J. (2018). Climate change and drought: From past to future. *Current Climate Change Reports*, *4*(2), 164–179. Retrieved from `https://doi.org/10.1007/s40641-018-0093-2` doi: 10.1007/s40641-018-0093-2

Cook, B. I., Mankin, J. S., Marvel, K., Williams, A. P., Smerdon, J. E., & Anchukaitis, K. J. (2020). Twenty-first century drought projections in the CMIP6 forcing scenarios. *Earth's Future*, *8*(6), e2019EF001461. Retrieved from `https://agupubs.onlinelibrary.wiley.com/doi/abs/10.1029/2019EF001461` (e2019EF001461 10.1029/2019EF001461) doi: https://doi.org/10.1029/2019EF001461

Danabasoglu, G., Lamarque, J.-F., Bacmeister, J., Bailey, D. A., DuVivier, A. K., Edwards, J., ... Strand, W. G. (2020). The Community Earth System Model Version 2 (CESM2). *Journal of Advances in Modeling Earth Systems*, *12*(2), e2019MS001916. Retrieved from `https://agupubs.onlinelibrary.wiley.com/doi/abs/10.1029/2019MS001916` (e2019MS001916 2019MS001916) doi: https://doi.org/10.1029/2019MS001916

Deser, C., Lehner, F., Rodgers, K. B., Ault, T., Delworth, T. L., DiNezio, P. N., ... Ting, M. (2020). Insights from earth system model initial-condition large ensembles and future prospects. *Nature Climate Change*, *10*(4), 277–

286. Retrieved from https://doi.org/10.1038/s41558-020-0731-2 doi: 10.1038/s41558-020-0731-2

Dettinger, M., Udall, B., & Georgakakos, A. (2015). Western water and climate change. *Ecological Applications*, *25*(8), 2069-2093. Retrieved from https://esajournals.onlinelibrary.wiley.com/doi/abs/10.1890/15-0938.1 doi: https://doi.org/10.1890/15-0938.1

Duan, S. Q., McKinnon, K. A., & Simpson, I. R. (2025). The impact of soil preconditioning on the evolution of heatwaves under constrained circulation: A case study of the 2021 pacific northwest heatwave. *Earth's Future*, *13*(9), e2025EF006216. Retrieved from https://agupubs.onlinelibrary.wiley.com/doi/abs/10.1029/2025EF006216 (e2025EF006216 2025EF006216) doi: https://doi.org/10.1029/2025EF006216

Eyring, V., Bony, S., Meehl, G. A., Senior, C. A., Stevens, B., Stouffer, R. J., & Taylor, K. E. (2016). Overview of the coupled model intercomparison project phase 6 (cmip6) experimental design and organization. *Geoscientific Model Development*, *9*(5), 1937–1958. Retrieved from https://gmd.copernicus.org/articles/9/1937/2016/ doi: 10.5194/gmd-9-1937-2016

Fang, Z., Zhang, W., Brandt, M., Abdi, A. M., & Fensholt, R. (2022). Globally increasing atmospheric aridity over the 21st century. *Earth's Future*, *10*(10), e2022EF003019. Retrieved from https://agupubs.onlinelibrary.wiley.com/doi/abs/10.1029/2022EF003019 (e2022EF003019 2022EF003019) doi: https://doi.org/10.1029/2022EF003019

Fischer, E. M., Beyerle, U., Bloin-Wibe, L., Gessner, C., Humphrey, V., Lehner, F., ... Knutti, R. (2023). Storylines for unprecedented heatwaves based on ensemble boosting. *Nature Communications*, *14*(1), 4643. Retrieved from https://doi.org/10.1038/s41467-023-40112-4 doi: 10.1038/s41467-023-40112-4

Gessner, C., Fischer, E. M., Beyerle, U., & Knutti, R. (2021). Very rare heat extremes: Quantifying and understanding using ensemble reinitialization. *Journal of Climate*, *34*(16), 6619 - 6634. Retrieved from https://journals.ametsoc.org/view/journals/clim/34/16/JCLI-D-20-0916.1.xml doi: 10.1175/JCLI-D-20-0916.1

Hersbach, H., Bell, B., Berrisford, P., Hirahara, S., Horányi, A., Muñoz-Sabater, J., ... Thépaut, J.-N. (2020). The era5 global reanalysis. *Quarterly Journal of the Royal Meteorological Society*, *146*(730), 1999-2049. Retrieved from https://rmets.onlinelibrary.wiley.com/doi/abs/10.1002/qj.3803 doi: https://doi.org/10.1002/qj.3803

Higuera, P. E., & Abatzoglou, J. T. (2021). Record-setting climate enabled the extraordinary 2020 fire season in the western united states. *Global Change Biology*, *27*(1), 1-2. Retrieved from https://onlinelibrary.wiley.com/doi/abs/10.1111/gcb.15388 doi: https://doi.org/10.1111/gcb.15388

Hirschi, M., Seneviratne, S. I., Alexandrov, V., Boberg, F., Boroneant, C., Christensen, O. B., ... Stepanek, P. (2011). Observational evidence for soil-moisture impact on hot extremes in southeastern europe. *Nature Geoscience*, *4*(1), 17–21. Retrieved from https://doi.org/10.1038/ngeo1032 doi: 10.1038/ngeo1032

Hoell, A., Quan, X.-W., Hoerling, M., Fu, R., Mankin, J., Simpson, I., ... Sheffield, A. (2022). Record low north american monsoon rainfall in 2020 reignites drought over the american southwest. *Bulletin of the American Meteorological Society*, *103*(3), S26 - S32. Retrieved from https://journals.ametsoc.org/view/journals/bams/103/3/BAMS-D-21-0129.1.xml doi: 10.1175/BAMS-D-21-0129.1

Holden, Z. A., Swanson, A. K., Sadegh, M., Luce, C. H., Noonan-Wright, E., & Parsons, R. A. (2025). Soil moisture is a stronger predictor of forest fire spread potential than weather in the u.s. northern rocky mountains. *Geophysical Research Letters*, *52*(16), e2025GL116248. Retrieved from https://

agupubs.onlinelibrary.wiley.com/doi/abs/10.1029/2025GL116248 (e2025GL116248 2025GL116248) doi: https://doi.org/10.1029/2025GL116248

Huang, B., Thorne, P. W., Banzon, V. F., Boyer, T., Chepurin, G., Lawrimore, J. H., ... Zhang, H.-M. (2017). Extended reconstructed sea surface temperature, version 5 (ersstv5): Upgrades, validations, and intercomparisons. *Journal of Climate*, *30*(20), 8179 - 8205. Retrieved from https://journals.ametsoc.org/view/journals/clim/30/20/jcli-d-16-0836.1.xml doi: 10.1175/JCLI-D-16-0836.1

Jiang, Y., Smerdon, J. E., Seager, R., Wang, G., Cook, B. I., Zheng, C., ... Williams, A. P. (2025). The influence of land-surface conditions on the 2020–2021 western us drought. *Water Resources Research*, *61*(5), e2024WR038124. Retrieved from https://agupubs.onlinelibrary.wiley.com/doi/abs/10.1029/2024WR038124 (e2024WR038124 2024WR038124) doi: https://doi.org/10.1029/2024WR038124

Kumar, S., Merwade, V., Kinter, J. L., & Niyogi, D. (2013). Evaluation of temperature and precipitation trends and long-term persistence in cmip5 twentieth-century climate simulations. *Journal of Climate*, *26*(12), 4168 - 4185. Retrieved from https://journals.ametsoc.org/view/journals/clim/26/12/jcli-d-12-00259.1.xml doi: 10.1175/JCLI-D-12-00259.1

Mankin, J. S., Simpson, I. R., Hoell, A., Fu, R., Lisonbee, J., Sheffield, A., & Barrie, D. (2021). *NOAA drought task force report on the 2020–2021 Southwestern U.S. drought* (Tech. Rep.). NOAA Drought Task Force IV.;National Integrated Drought Information System (U.S.);Modeling, Analysis, Predictions, and Projections Program (U.S.). doi: https://doi.org/10.25923/j0rj-an85

Newman, R., & Noy, I. (2023). The global costs of extreme weather that are attributable to climate change. *Nature Communications*, *14*(1), 6103. Retrieved from https://doi.org/10.1038/s41467-023-41888-1 doi: 10.1038/s41467-023-41888-1

Papalexiou, S. M., Rajulapati, C. R., Clark, M. P., & Lehner, F. (2020). Robustness of cmip6 historical global mean temperature simulations: Trends, long-term persistence, autocorrelation, and distributional shape. *Earth's Future*, *8*(10), e2020EF001667. Retrieved from https://agupubs.onlinelibrary.wiley.com/doi/abs/10.1029/2020EF001667 (e2020EF001667 2020EF001667) doi: https://doi.org/10.1029/2020EF001667

Pascale, S., Boos, W. R., Bordoni, S., Delworth, T. L., Kapnick, S. B., Murakami, H., ... Zhang, W. (2017). Weakening of the north american monsoon with global warming. *Nature Climate Change*, *7*(11), 806–812. Retrieved from https://doi.org/10.1038/nclimate3412 doi: 10.1038/nclimate3412

Pendergrass, A. G., Knutti, R., Lehner, F., Deser, C., & Sanderson, B. M. (2017). Precipitation variability increases in a warmer climate. *Scientific Reports*, *7*, 1.

Rodgers, K. B., Lee, S.-S., Rosenbloom, N., Timmermann, A., Danabasoglu, G., Deser, C., ... Yeager, S. G. (2021). Ubiquity of human-induced changes in climate variability. *Earth System Dynamics*, *12*(4), 1393–1411. Retrieved from https://esd.copernicus.org/articles/12/1393/2021/ doi: 10.5194/esd-12-1393-2021

Seager, R., Ting, M., Alexander, P., Nakamura, J., Liu, H., Li, C., & Simpson, I. R. (2022). Mechanisms of a meteorological drought onset: Summer 2020 to spring 2021 in southwestern north america. *Journal of Climate*, *35*(22), 7367 - 7385. Retrieved from https://journals.ametsoc.org/view/journals/clim/35/22/JCLI-D-22-0314.1.xml doi: 10.1175/JCLI-D-22-0314.1

Shepherd, T. G. (2014). Atmospheric circulation as a source of uncertainty in climate change projections. *Nature Geoscience*, *7*(10), 703–708. Retrieved from https://doi.org/10.1038/ngeo2253 doi: 10.1038/ngeo2253

Simpson, I. R., McKinnon, K. A., Kennedy, D., Lawrence, D. M., Lehner, F., &

Seager, R. (2024). Observed humidity trends in dry regions contradict climate models. *Proceedings of the National Academy of Sciences*, *121*(1), e2302480120. Retrieved from https://www.pnas.org/doi/abs/10.1073/pnas.2302480120 doi: 10.1073/pnas.2302480120

Wieder, W. R., Kennedy, D., Lehner, F., Musselman, K. N., Rodgers, K. B., Rosenbloom, N., ... Yamaguchi, R. (2022). Pervasive alterations to snow-dominated ecosystem functions under climate change. *Proceedings of the National Academy of Sciences*, *119*(30), e2202393119. Retrieved from https://www.pnas.org/doi/abs/10.1073/pnas.2202393119 doi: 10.1073/pnas.2202393119

Williams, A. P., Cook, B. I., & Smerdon, J. E. (2022). Rapid intensification of the emerging southwestern north american megadrought in 2020–2021. *Nature Climate Change*, *12*(3), 232–234. Retrieved from https://doi.org/10.1038/s41558-022-01290-z doi: 10.1038/s41558-022-01290-z

Wu, Y., Miao, C., Slater, L., Fan, X., Chai, Y., & Sorooshian, S. (2024). Hydrological projections under cmip5 and cmip6: Sources and magnitudes of uncertainty. *Bulletin of the American Meteorological Society*, *105*(1), E59 - E74. Retrieved from https://journals.ametsoc.org/view/journals/bams/105/1/BAMS-D-23-0104.1.xml doi: 10.1175/BAMS-D-23-0104.1